13431 NE 20th St,
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SEA·BIRD SCIENTIFIC

(425) 643-9866
www.seabird.com

ECO Calibration and Repairs

Diagnosis: Evaluated instrument (FLBBCDSLC-6982) It was found that the unit required tuning.

Repairs and Modifications: The unit was tuned, calibrated, and standard service was performed. Unit returned with customer settings [11-14 LP].

Comments: New faceplate added. New device files and characterization sheets included.

ECO Standard Service:

The instrument's bulkhead connector, pressure housing and optical head are inspected for damage. The optics are inspected for any crack and/ or delamination. The instrument is also evaluated to determine proper functionality.

If applicable, the head is inspected for cracks in motor bores. Case seals, shaft, shaft seal, faceplate, wiper, desiccant pack, and batteries (if equipped) are replaced.

Incoming settings and memory are collected if incoming conditions allow. When requested, a pre-service characterization can be performed. Then, the final calibration and characterization are conducted. Noise, stability, and live pressure test are also performed before service is completed.

A device file, repair sheet, and new characterization sheets are provided to customers via USB.

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Scattering Meter Calibration Sheet

11/14/2025
Wavelength: 700 S/N FLBBCDSLC-6982

Use the following equation to obtain "scaled" output values:

$$\beta(\theta_c)\ m^{-1}\ sr^{-1} = \text{Scale Factor} \times (\text{Output} - \text{Dark Counts})$$

• **Scale Factor for 700 nm**	=	1.828E-06	$(m^{-1}sr^{-1})$/counts	
• **Output**	=	meter reading	counts	
• **Dark Counts**	=	49	counts	
Instrument Resolution	=	1.1	counts	2.07E-06 $(m^{-1}sr^{-1})$

Definitions:

- **Scale Factor**: Calibration scale factor, $\beta(\theta_c)$/counts. Refer to User's Guide for derivation.
- **Output**: Measured signal output of the scattering meter.
- **Dark Counts**: Signal obtained by covering detector with black tape and submersing sensor in water.

Instrument Resolution: Standard deviation of 1 minute of collected data.

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ECO CDOM Fluorometer Characterization Sheet

Date: 11/14/2025

S/N: FLBBCDSLC-6982

CDOM concentration expressed in ppb can be derived using the equation:

CDOM (ppb) = Scale Factor * (Output - Dark Counts)

	Digital
Dark Counts	50 counts
Scale Factor (SF)	0.0907 ppb/count
Maximum Output	4130 counts
Resolution	1.0 counts
Ambient temperature during characterization	22.0 °C

Dark Counts: Signal output of the meter in clean water with black tape over detector.

SF: Determined using the following equation: SF = x ÷ (output - dark counts), where x is the concentration of the solution used during instrument characterization. SF is used to derive instrument output concentration from the raw signal output of the fluorometer.

Maximum Output: Maximum signal output the fluorometer is capable of.

Resolution: Standard deviation of 1 minute of collected data.

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ECO Chlorophyll Fluorometer Characterization Sheet

Date: 11/14/2025

S/N: FLBBCDSLC-6982

Chlorophyll concentration expressed in µg/l can be derived using the equation:

CHL (µg/l) = Scale Factor * (Output - Dark counts)

	Digital
Dark counts	49 counts
Scale Factor (SF)	0.0073 µg/l/count
Maximum Output	4130 counts
Resolution	1.0 counts
Ambient temperature during characterization	22.0 °C

Dark Counts: Signal output of the meter in clean water with black tape over detector.

SF: Determined using the following equation: SF = x ÷ (output - dark counts), where x is the concentration of the solution used during instrument characterization. SF is used to derive instrument output concentration from the raw signal output of the fluorometer.

Maximum Output: Maximum signal output the fluorometer is capable of.

Resolution: Standard deviation of 1 minute of collected data.

The relationship between fluorescence and chlorophyll-a concentrations in-situ is highly variable. The scale factor listed on this document was determined using a mono-culture of phytoplankton (Thalassiosira weissflogii). The population was assumed to be reasonably healthy and the concentration was determined by using the absorption method. To accurately determine chlorophyll concentration using a fluorometer, you must perform secondary measurements on the populations of interest. This is typically done using extraction-based measurement techniques on discrete samples. For additional information on determining chlorophyll concentration see "Standard Methods for the Examination of Water and Wastewater" part 10200 H, published jointly by the American Public Health Association, American Water Works Association, and the Water Environment Federation.